

RESEARCH ARTICLE

Glutamate-Weighted CEST Contrast After Removal of Magnetization Transfer Effect in Human Brain and Rat Brain with Tumor

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Abstract

Purpose: To mitigate the effect of magnetization transfer (MT) from glutamate-weighted chemical exchange saturation transfer (GluCEST) contrast in healthy human brain and its demonstration in a rat brain tumor model.

Procedures: GluCEST data was acquired from six healthy human volunteers at 7T and on a rat brain with tumor at 9.4 T. Single voxel proton magnetic resonance spectroscopy (¹HMRs) data was acquired from three human volunteers. The magnetic resonance imaging protocol included CEST data acquisition at multiple frequencies for generating Z-spectra, B₀ and B₁ map. Partial Z-spectra at offset frequencies from ±100 to ±14 ppm were fitted to model semi-solid MT component by Lorentzian, Gaussian, super-Lorentzian, and 6th degree polynomial function lineshapes. Average residual errors per pixel was calculated. The MT effect of the Z-spectra was removed by subtracting fitted MT component from Z-spectra. GluCEST was computed as GluCEST_{Neg} (normalized with signal at −3 ppm) and GluCEST_{M0} (normalized with unsaturated signal). The difference between GluCEST maps before and after MT removal was compared using *T* test.

Results: Better accuracy of fitting off-resonance Z-spectra was achieved with super-Lorentzian ($\sigma=0.0009$) and Lorentzian ($\sigma=0.0017$) compared to other lineshapes. There was significant ($p<0.01$) increase in GluCEST_{M0} and decrease in GluCEST_{Neg} contrast after MT removal. GluCEST_{Neg} and GluCEST_{M0} maps after MT removal using Lorentzian lineshape showed gray matter (GM) to white matter (WM) contrast ratio of 1.47 and 1.52 respectively. These ratios are close to glutamate concentration ratio in GM/WM as observed from ¹HMRs data. Thus, the quantity of the MT removed from Z-spectra is appropriate using Lorentzian lineshape due to preservation of GluCEST contrast-ratio in GM/WM. The amount of MT removed from Z-spectra is overestimated using super-Lorentzian and underestimated using Gaussian and polynomial lineshapes. Tumor tissue showed unexpected increase in GluCEST contrast compared to contra-lesional tissue, which represents normal appearing tissue in the brain on contra-lateral side of tumor region, due to decrease in MT component.

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Conclusions: Removal of MT effect from Z-spectra using Lorentzian lineshape increased the specificity of GluCEST contrast to glutamate in healthy human brain and was demonstrated in rat brain tumor model.

Key words: Magnetic resonance imaging (MRI), Chemical exchange saturation transfer (CEST), Glutamate, GluCEST, Z-spectra, Magnetization transfer, Lorentzian, Super-Lorentzian

Introduction

Glutamate is an excitatory neurotransmitter that originates from nerve cells of the central nervous system (CNS). Alterations in glutamate levels have been observed in various neurological disorders [1–3]. Early assessment of changes in glutamate concentrations and its detection are very crucial for better diagnosis of disease. Currently, single voxel proton magnetic resonance spectroscopy (^1HMR S) [1, 4–7] and magnetic resonance spectroscopic imaging (MRSI) [8] have been widely used for non-invasive *in vivo* detection of glutamate concentrations. However, glutamate estimation using spectroscopy suffers from several limitations such as low spatial resolution and estimation of glutamate peaks that are quite difficult to resolve due to overlap with other brain metabolites. Positron emission tomography (PET) imaging is another widely used technique to detect glutamate concentrations [9]. The major drawback of the PET technique lies in use of radio-tracer elements and it suffers from low spatial resolution. Recently, Cai et al. [10] demonstrated an advanced non-invasive magnetic resonance imaging (MRI) technique based on chemical exchange saturation transfer (CEST) for *in vivo* molecular mapping of glutamate with higher spatial and temporal resolution along with higher sensitivity when compared to ^1HMR S. Several studies were carried with glutamate-weighted CEST (GluCEST) imaging and it shows promise *in vivo* for both preclinical and clinical applications as demonstrated in neuro-pathological disorders [11], epilepsy [12], psychosis [13], brain tumors [14], Huntington's disease [15] and Alzheimer's disease (AD) [16], brain abscess [17], and sleep-related psychological disorders [18].

GluCEST contrast is generally computed using CEST asymmetry (CEST_{asy}) analysis [10, 11, 14, 16, 17, 19–22] which is also widely known as magnetization transfer ratio asymmetry (MTR_{asy}). MTR_{asy} is measurement metric. It has been well reported [23–25] that magnetization transfer (MT) effect from semi-solid macromolecules is asymmetric. However, its effect is expected to underestimate any CEST contrast computed using asymmetry analysis, including GluCEST. Although asymmetry in MT effect is well known, its contribution close to on-resonance frequencies is poorly reported for GluCEST. Therefore, there is a need for a systematic study to estimate MT asymmetry contribution as well as methods for its removal from GluCEST contrast. In general, CEST data can be acquired over a large range, such as Z-spectra, or in a limited range depending upon time and

method of CEST contrast computation. Z-spectral fitting is an alternate approach to CEST_{asy} which has been developed to extract individual components of Z-spectra without any effect from MT, MT asymmetry. However, to the best of our knowledge, Z-spectral-based fitting has not yet been reported for computing GluCEST contrast probably due to high saturation B_1 power used. Some of the studies [26, 27] have utilized partial Z-spectral data from off-resonance frequencies to model MT effect. In this study, we plan to use the same approach to model MT effect and remove its contribution from GluCEST weighted data before computing actual GluCEST contrast.

Several methods are available to model broad MT spectrum using different lineshapes and to remove MT effect from Z-spectrum. A study [28] reported the use of two pool modified Bloch McConnell equation with exchange rates to model MT pool. The Bloch equation assumes Lorentzian lineshape for MT pool, but narrow Lorentzian lineshape is inappropriate to fit broad MT spectrum [29]. Iino et al. [30] reported that the immobile spins of semi-solid macromolecules are coupled through dipole-dipole Hamiltonian and thus, the line width of the macromolecules is much wider. It was assumed that the best fits for liquid and solid pools were Lorentzian and Gaussian lineshape, respectively. Since MT is a semi-solid pool, it can be modeled by a mixture of Lorentzian and Gaussian lineshape. Morrison et al. [31] incorporated a non-Bloch behavior into mathematical modeling of MT pool and reported that super-Lorentzian lineshape provides better approximation to fit MT pool. Quesson et al. [32] also utilized super-Lorentzian lineshape for MT pools to investigate rat brains at 4.7 T. Also, in one of the pilot studies [33], super-Lorentzian lineshape has been reported to be the best fit to model broad MT effect in the Z-spectra of agar solution and biological tissues with steady state condition when compared to Lorentzian and Gaussian fits. On the contrary, Gaussian lineshape gave better approximation than super-Lorentzian to model MT [34]. Singh et al. [27] modeled semi-solid MT pool using polynomial equations of 6th degree. Apparent exchange-dependent relaxation method (AREXfit) [35, 36] along with fitting approach can remove contaminations from MT and rNOE [37]. The absorption lineshape for MT depends on various factors such as (i) compositions of the macromolecular tissues, (ii) experimental conditions (steady/non-steady state), (iii) saturation power (B_1) used, (iv) continuous wave or transient irradiation pulse, and (v) shape of the saturation pulse. Considering all the cases, it is better

not to use *a priori* lineshape [38, 39], but rather use a best fitted lineshape to the given Z-spectra for MT modeling. Cai et al. [8] reported ratio of glutamate concentration as well as GluCEST between GM and WM. This study reported a strong correlation between two approaches. While choosing MT lineshape, one should also consider the correlation of GluCEST ratio and glutamate concentrations between GM and WM. Therefore, depending upon type of Z-spectral data and saturation parameters, appropriate lineshape should be used to preserve the contrast ratio after MT removal.

A previous study [10] reported an increase in GluCEST contrast in tumor tissue of rat brain when compared to contra-lateral side tissue. This increase in GluCEST contrast could be due to decrease in tissue pH as well as an increase in concentration of Glutamate in tumor tissue [40]. Moreover, it has been well reported [41] that there is a substantial decrease in MT component in tumor tissue when compared to contra-lateral side tissue. An increase in conventional CEST_{asy} contrast could be due to true changes in GluCEST contrast as well as a decrease in MT effect. So, there is a need to study changes in GluCEST contrast in the tumor tissue after MT removal.

The first objective of this study was therefore to model the broad semi-solid MT pool using different lineshapes as well as to investigate the best lineshape to approximate MT pool for the given set of saturation pulse parameters. The second objective being the estimation of GluCEST contrast from *in vivo* human brain after removal of MT contribution from Z-spectra acquired at 7T. In addition, a preliminary study was carried out to evaluate effect of MT removal on GluCEST contrast in tumor tissue of rat brain at 9.4T.

Methods

Human Studies

Six asymptomatic human volunteers (M:F=5:1; 35 ± 14.5 years) were recruited in this study with written informed consents obtained from each volunteer. All the studies were conducted under an approved Institutional Review Board protocol at the University of Pennsylvania. MR imaging was performed on 7T Siemens research scanner (Siemens Medical Solutions, PA) using a 32 channel receive and single channel transmit proton phased array head coil. Proper cushions and pads were used for reducing head movements of the volunteers during the MRI scan.

Acquisition Protocol for Human MRI Scans

The MRI study protocol consisted of the following steps: (i) a three-plane localizer, (ii) WASSR [42] data, (iii) B₁ map data, (iv) T₁-weighted imaging, (v) CEST or Z-spectra data (using a pulse sequence reported previously [10, 26]), and (vi) a control image without saturation. The saturation pulse train of CEST sequence consisted of a series of Hanning

windowed rectangular pulses of 100 ms duration each separated by a 200 μs delay (duty cycle = 99.80 %). WASSR data were acquired over the frequency range of -1 to +1 ppm with step size of 0.1 ppm with saturation pulse of B_{1rms} = 0.29 μT and with 200 ms pulse duration. The 2D CEST MRI imaging parameters were slice thickness = 5 mm, gradient readout echo (GRE) flip angle (FA) = 6°, time of repetition (TR) / time of echo (TE) = 6/1.4 ms, field of view (FOV) = 240 × 240 mm², matrix size = 240 × 180, and shot TR = 6 s. Z-spectra data were acquired using saturation pulse of root mean square B₁ (B_{1rms}) of 0.7, 1.4, 2.2, and 2.9 μT and with 800 ms pulse duration (series of 100 ms pulse train) at the following offset frequencies: ±0 to ±5 ppm in steps of 0.1 ppm, ±6 to ±10 ppm in steps of 1 ppm, ±12 to ±20 ppm in steps of 2 ppm, ±30 to ±100 ppm in steps of 5 ppm.

Acquisition Protocol for Human MR Spectroscopy

¹H MRS was also carried out for three out of six healthy human volunteers on the same slice location in different imaging session. For selecting same slice location for spectroscopy data as used in CEST data, we used a co-registration program, ImScribe (accessible at <https://www.med.upenn.edu/cmroi/imscribe.html>) and procedure described in Nanga et al. [43]. The spectra were acquired for voxels positioned on GM and WM tissues as shown in Fig. 5b and c using point resolved spectroscopy (PRESS) sequence with the following parameters: voxel dimensions = 40 × 10 × 5 mm³, TR = 3000 ms, TE = 23 ms, number of averages for water reference/water suppressed spectra = 8/64, spectral width = 4 kHz, number of points = 2048, number of dummy scans = 4. The total acquisition time for spectroscopy on each of the voxels was 5 min. Localized shimming was performed to obtain water line widths of 0.08 ppm or less. Water suppression was done with variable pulse power and optimization relaxation (VAPOR) pulse cluster pre-encoded to PRESS sequence.

The acquired spectra were analyzed using LC model which calculates the best fit of the experimental spectrum as a linear combination of basis spectra. The LC model fit was performed over the spectral range between 0.5 and 4.0 ppm and the peak integrals for all metabolites were calculated from the fitted peaks. Unsuppressed water spectrum obtained for each voxel of interest was used in the LC Model to compute the concentration of glutamate and other metabolites.

Animal Preparation and Experiment

Animal studies were performed according to an approved Institutional Animal Care and Use Committee (IACUC) protocol at the University of Pennsylvania. To develop intracranial tumors, rat gliosarcoma cells (9 L) were injected on a Syngeneic female Fisher rats (F344/NCR, 4–6 weeks old)

weighing 130–150 g. Rat was subjected to MRI 5 weeks after implantation of tumor cells. Rat was scanned in 9.4T horizontal bore small animal scanner (Varian, Palo Alto, CA) and placed inside a 35-mm-diameter commercial quadrature proton coil (m2 m Imaging Corp., Cleveland, OH). Rat was anesthetized using 1.5 % isoflurane in 1 l/min oxygen followed by intra-peritoneal injection of a ketamine (91 mg/kg) and acepromazine (9.1 mg/kg) mixture. Rat was kept warm with warm air generated from a heater (SA Instruments, Inc., Stony Brook, NY). Respiration and body temperature were monitored using MRI compatible small animal monitor system (SA Instruments, Inc., Stony Brook, NY).

Imaging Protocol for In Vivo Animal Study

CEST imaging of tumor-bearing rat brain was performed using the following sequence parameters: FOV = 35×35 mm², slice thickness = 2 mm, GRE FA = 15°, GRE readout TR = 6.2 ms, TE = 2.9 ms, matrix size = 128×128 , number of averages = 2, and one saturation pulse with duration = 1 s and root mean squared B₁ (B_{1rms}) = 5.9 μT (250 Hz) followed by a 32 segment acquisition every 4 s. CEST images were collected at multiple frequencies (ppm) $\pm 100, \pm 50, \pm 25, \pm 12$ to ± 6 with step size of 2, ± 5 to ± 0 with step size of 0.25. B₀ maps were generated using WASSR [42] data.

B₀ Inhomogeneity Correction

B₀ map was generated from WASSR data followed by B₀ inhomogeneity correction of Z-spectral data [42]. For B₀ inhomogeneity correction, each voxel data at offset frequency ($\Delta\omega$ ppm) was replaced by the fitted data from the frequency shift in the B₀ map ($\Delta\omega - \delta\omega_0$) ppm, where $\delta\omega_0$ is obtained from B₀ map at that corresponding voxel.

GluCEST Contrast Computation

B₀ corrected Z-spectral data were generated by the following equation:

$$Z\text{-spectrum}(\Delta\omega) = \frac{M_{\text{sat}}(\Delta\omega)}{M_0} \quad (1)$$

GluCEST map was finally computed from Z-spectra at B_{1rms} of 2.9 μT using the following Equations (2) and (3):

$$\text{GluCEST}_{\text{Neg}} = \frac{M_{\text{sat}}(-\Delta\omega) - M_{\text{sat}}(+\Delta\omega)}{M_{\text{sat}}(-\Delta\omega)} \times 100 \quad (2)$$

$$\text{GluCEST}_{\text{M0}} = \frac{M_{\text{sat}}(-\Delta\omega) - M_{\text{sat}}(+\Delta\omega)}{M_0} \times 100 \quad (3)$$

where $M_{\text{sat}}(+\Delta\omega)$ and $M_{\text{sat}}(-\Delta\omega)$ are the signal intensities (SI) with ($\Delta\omega = 3$ ppm) downfield and upfield with respect to water resonance. M_0 is the SI without RF saturation.

B₁ Inhomogeneity Correction

The current study used three flip angles (20°, 40°, and 80°) and double angle method [10] to generate transmit B₁ field map. The final B₁ map was generated as an average of two B₁ maps computed using flip angles (20°, 40°) and (40°, 80°). The GluCEST map was further corrected for B₁ inhomogeneity using calibration method as described in the previous study [19]. In the current study, we have collected data at four different B_{1rms} to compute the calibrations parameters for gray matter (GM) and white matter (WM) mask.

MT Modeling by Different Lineshapes: Fitting Functions

MT effect dominates beyond ± 14 ppm range in Z-spectra at 7T with negligible contributions from CEST, nuclear Overhauser effect (NOE) and direct saturation (DS). Partial Z-spectra at far-off resonance offset frequencies (± 100 to ± 14 ppm) was fitted with semi-solid MT component with each one of the lineshapes—Lorentzian (equation iv), Gaussian (equation v), and super-Lorentzian (Eq. 6), and the 6th degree polynomial function. The equations for the Lorentzian, Gaussian, and super-Lorentzian lineshapes were described in Morrison et al. [31, 33] which have been modified (as shown in Eqs. iv, v, and vi) and was used in this current study.

$$\text{Lorentzian} : g_i(2\pi(w-w_0)) = \frac{1}{[1 + \{2\pi(w-w_0) * T_{2m}\}^2]} \times A \quad (4)$$

$$\text{Gaussian} : g_i(2\pi(w-w_0)) = e^{-\frac{\{2\pi(w-w_0) * T_{2m}\}^2}{2}} \times A \quad (5)$$

super-Lorentzian:

$$g_i(2\pi(w-w_0)) = \int_0^{\pi/2} \frac{\sin\theta}{|3(\cos\theta)^2 - 1|} \times e^{-2 \times \left(\frac{2\pi(w-w_0) * T_{2m}}{|3(\cos\theta)^2 - 1|} \right)^2} d\theta \times A \quad (6)$$

where w is the frequency offset from the water resonance, w_0 is the offset frequency of the MT pool, θ is the dipolar Hamiltonian angle [33], and A is the scaling factor. The center of MT pool varies across tissues from -1 to -2.5 ppm as reported in various studies [44, 45] and thus, we provided larger range between upper and lower bound for fitting (Table 3). T_{2m} is time constant which varies for different lineshapes and is not to be confused with transverse relaxation time (T_2) of the MT pool [39]. T_{2m} is not the true T_2 of the MT pool since we are fitting Z-spectra data not spectroscopic data. T_{2m} characterizes the linewidth of the solid macromolecular MT pool [33]. T_{2m} , w_0 , and A are the parameters of the fit.

After partial fitting of Z-spectra, the fitted MT component was interpolated over the entire offset frequency range (-100 to $+100$ ppm) using the fitted parameters of the respective lineshape. There is an exception to this interpolation in case of super-Lorentzian lineshape where a sudden hump at the center of MT pool was observed. Thus, for super-Lorentzian lineshape, spline interpolation was used to avoid the pole or hump [44]. The fitted MT component was then removed from original Z-spectra. GluCEST contrast was then computed at $+3$ ppm using Z-spectral data with and without MT removal.

Post-processing

All image processing and data analysis were performed using in-house developed programs written in MATLAB-R2017a (The MathWorks, Inc., Natick, MA, USA). For motion correction, all images (WASSR, B_0 and B_1 maps, CEST-weighted) were registered to the control image using affine transformation followed by manual de-scalping [46]. High-resolution T_1 -weighted image was used for segmentation of gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) using in-house developed MATLAB code. After segmentation, the WM mask and GM mask were further eroded using morphological operation to remove partial volume effect. Z-spectra were normalized with unsaturated signals ($\Delta w = 100$ ppm). Z-spectral fitting was performed using in-house written program that utilizes in-built nonlinear fitting “lsqcurvefit” routine of MATLAB. Parameters used as an initial guess and bound constraints are listed in Table 3. The average values of B_0 corrected GluCEST contrast for region of interest (ROI) were computed for tumor tissue as well as for contra-lateral normal appearing tissue. MTR was computed as $100 \times (1 - M_{\text{sat}(+20)} / M_{\text{sat}(+100)})$.

Statistical Analysis

For evaluating quality of fit, the average residual error per point (σ) was calculated for each fitting as described in Morison et al. [31]. Goodness of fitting (R^2) was not calculated in this study for a reason mentioned in the previously reported study [33]. The MT uncorrected and corrected GluCEST maps were compared using independent student's t test with two tails. The difference was considered statistically significant if $p < 0.05$.

Results

MT Effect Modeling

Figure 1 a–d shows partial fitting (± 100 to ± 14 ppm) of a Z-spectrum from a circular region of interest (ROI) (radius of 2 units) drawn on WM tissue of a representative healthy volunteer using Lorentzian, Gaussian, super-Lorentzian

lineshapes, and a 6th degree polynomial function. For far-off resonance partial data, fitting of the Z-spectrum corresponding to MT effect with Gaussian lineshape was less accurate ($\sigma = 0.0026$) compared to fitting using polynomial lineshape ($\sigma = 0.0019$), Lorentzian lineshape ($\sigma = 0.0017$), and super-Lorentzian lineshape ($\sigma = 0.0009$). The parameters of Z-spectral fitting for ROIs on GM and WM for the same volunteer are given in Table 1. Figure 1 e–h shows the interpolation of the partial Z-spectra fitted function over the whole offset frequency range. For the Lorentzian, Gaussian, and polynomial lineshapes, direct interpolation of the fitted function was used, whereas for super-Lorentzian lineshape, interpolation was carried out using spline function. The center and scaling factor of the fitted MT pool varies for different lineshapes (Table 1). The initial value for center of MT pool was fixed at -2.4 ppm (Table 3), fitted center for all the lineshapes varies between -1.3 and -2.2 ppm (Table 1). The values of interpolated function at water resonating frequency (0 ppm) are 0.54, 0.63, 0.56, and 0.63 for Lorentzian, Gaussian, super-Lorentzian, and polynomial lineshapes, respectively, which shows that the peak of MT pools varies for different lineshapes used in this study. The contribution of the MT component to the Z-spectrum varies with the selection of the lineshape. Since the amplitude of the MT pool is highest for super-Lorentzian, MT contribution to the Z-spectrum is highest considering this lineshape. Thus, the quantity of the MT component subtracted from Z-spectra varies with lineshape. Larger amount of MT was removed from Z-spectra using super-Lorentzian compared to other lineshapes. After the removal of MT component using super-Lorentzian lineshape, Z-spectra shifted higher compared to the shifts using other lineshapes.

Fig. S1 (see Electronic Supplementary Material (ESM)) shows fitting of MT component followed by its removal from Z-spectra generated from a circular ROI (radius of 3 units) over WM of a representative healthy human volunteer. The super-Lorentzian lineshape provided the best fit ($\sigma = 0.0009$). Lorentzian lineshape also fitted well ($\sigma = 0.0015$). Similar observations were noticed while fitting MT component of Z-spectra generated from an ROI over GM.

CEST Asymmetry Analysis

MT asymmetry (Fig. 2a–d) illustrates CEST_{asy} analysis before and after removal of MT component from Z-spectra generated from an ROI over GM. Figure 2 e–h shows CEST_{asy} analysis before and after removal of MT component from Z-spectra generated from an ROI over WM. The columns (left to right) represents MT modeling using Lorentzian, Gaussian, super-Lorentzian, and 6th degree polynomial. The CEST_{asy} contrast increases after removal of MT component from Z-spectra. This increase in CEST_{asy} is higher for super-Lorentzian lineshape followed by Lorentzian, Gaussian, and polynomial lineshapes in both

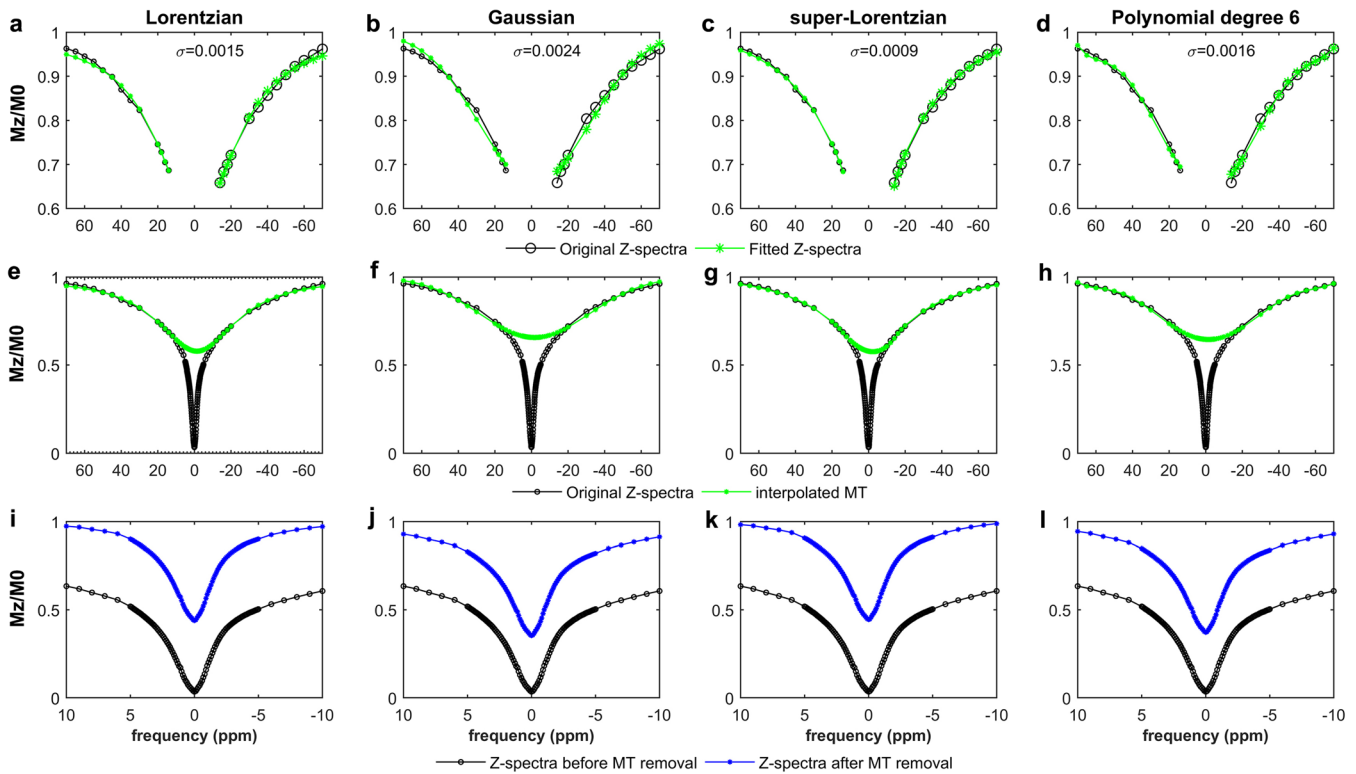


Fig. 1. Partial Z-spectra fitted with different lineshapes to model MT component and fitted MT component is removed from Z-spectra generated from an ROI drawn over GM. **a** Lorentzian, **b** Gaussian, **c** super-Lorentzian, and **d** 6th degree polynomial function lineshapes were used to fit partial Z-spectra (± 100 to ± 14 ppm). Fitted MT components were interpolated over the entire frequency range (-100 to $+100$ ppm) using **e** Lorentzian, **f** Gaussian, **g** spline, and **h** 6th degree polynomial functions. The fitted MT component was subtracted from Z-spectra to get modified Z-spectra without MT component (last row). For better visualization, -70 to $+70$ ppm data were plotted for first two rows and -10 to $+10$ ppm data were plotted in last row. MT magnetization transfer; GM gray matter; ROI regions of interest.

GM and WM. It has been observed that an increase in CEST_{asy} were similar for Gaussian and polynomial lineshapes. The increment of CEST_{asy} was more in WM than in GM which shows that more MT was being removed in WM compared with GM.

Table 1. Partial fitting of Z-spectra generated from an ROI drawn on GM and WM of a representative healthy human volunteer using different lineshapes to model broad MT spectrum

Tissue	Lineshape	T_{2m} (μs)	w_0 (ppm)	A (A.U.)	σ
GM	Lorentzian	20.21	-1.35	0.42	0.0015
	Gaussian	17.64	-1.65	0.34	0.0024
	Super-Lorentzian	6.39	-1.59	0.31	0.0009
	Polynomial degree 6	N/A	N/A	N/A	0.0016
WM	Lorentzian	22.25	-1.62	0.46	0.0017
	Gaussian	18.7	-2.14	0.36	0.0026
	Super-Lorentzian	6.97	-1.51	0.34	0.0009
	Polynomial degree 6	N/A	N/A	N/A	0.0019

ROI, regions of interest; GM, gray matter; WM, white matter; A, scaling factor; A.U., arbitrary unit; σ , average residual error per point; N/A, not applicable. T_{2m} is time constant which varies for different lineshapes and is not to be confused with transverse relaxation time (T_2) of the MT pool [39]. This T_{2m} characterizes the linewidth of the solid macromolecular MT pool [33]. w_0 is offset frequency of the MT pool which varies across tissues from -1 to -2.5 ppm as reported in various studies [44, 45]

GluCEST Maps Before and After MT Removal

The GluCEST_{M0} maps for different healthy subjects ($n=6$) before and after MT removal are shown in Fig. 3. The GluCEST_{M0} contrast was higher in GM compared with WM both without and with MT removal. The change in GluCEST_{M0} contrast was highest in super-Lorentzian followed by other lineshapes in both GM and WM. The GluCEST_{M0} contrast increased significantly ($p < 0.01$) from 3.17 ± 0.38 to 4.17 ± 0.38 % in GM and 1.22 ± 0.34 to 3.14 ± 0.31 % in WM considering super-Lorentzian lineshape. The GluCEST_{M0} contrast also increased significantly ($p < 0.01$) from 3.17 ± 0.38 to 3.87 ± 0.35 % in GM and 1.22 ± 0.34 to 2.64 ± 0.28 % in WM using Lorentzian lineshape. The GluCEST_{M0} contrast also increased using other two lineshapes but the difference between GluCEST_{M0} contrast before and after MT removal were not significant in GM, whereas there was significant change in WM (Table 2). The change in GluCEST_{M0} contrast was more in WM compared with GM after MT removal irrespective of any lineshape. Before MT removal, GluCEST_{M0} showed contrast ratio of GM/WM = 2.59, whereas, after MT removal, GluCEST_{M0} showed contrast ratio 1.47, 1.87, 1.32, and 1.88 for Lorentzian, Gaussian, super-Lorentzian, and 6th degree polynomial respectively. Table 4 presents the

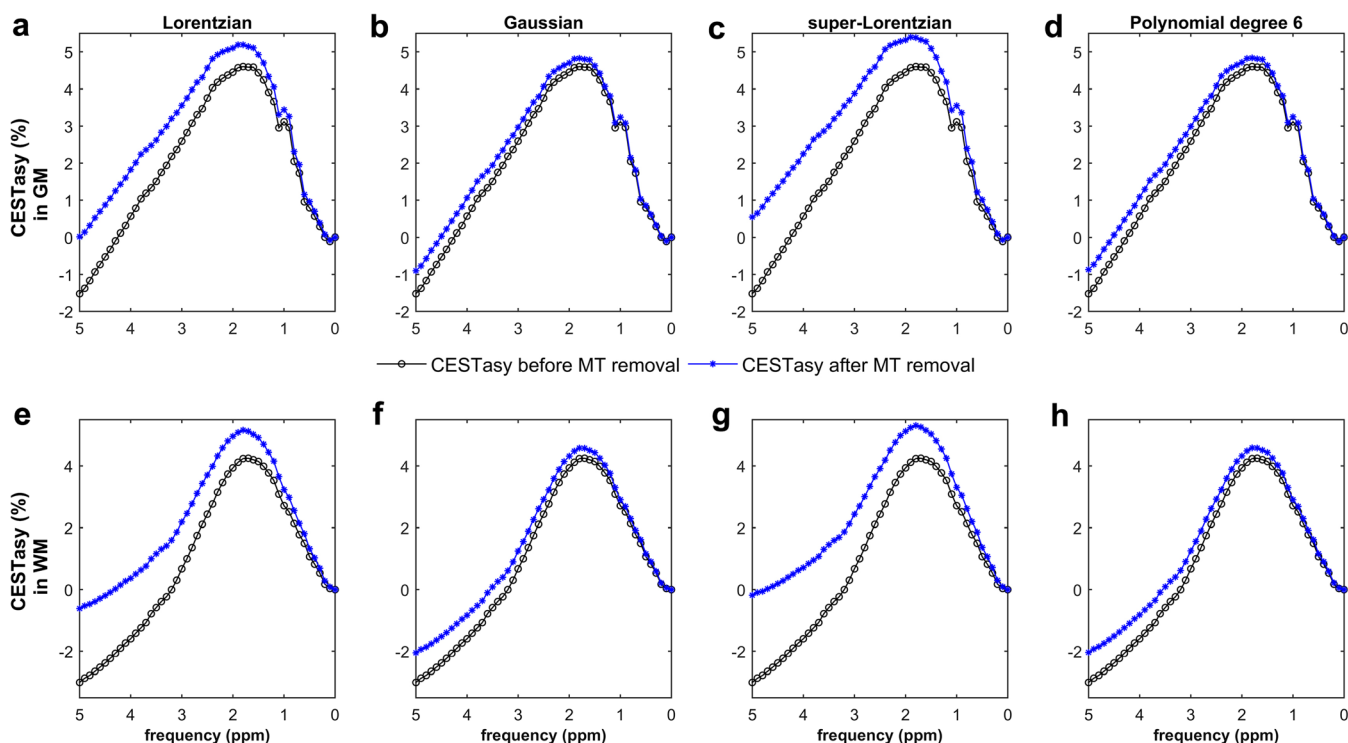


Fig. 2. Variation in CEST_{asy} over offset frequencies in **a–d** GM and **e–h** WM before and after MT removal. The MT component was modeled and fitted using **a, e** Lorentzian, **b, f** Gaussian, **c, g** super-Lorentzian with spline interpolation, and **d, h** 6th degree polynomial function lineshapes. CEST_{asy} increases after removal of MT component. For better visualization, 0 to 5 ppm data were plotted. Abbreviation: CEST_{asy} , CEST asymmetry analysis; MT, magnetization transfer; WM, white matter; GM, gray matter.

$\text{GluCEST}_{\text{Neg}}$ and $\text{GluCEST}_{\text{M0}}$ contrast GM/WM ratio before and after MT removal using different lineshapes considering only those three subjects that was used for spectroscopic study.

Figure 4 shows the $\text{GluCEST}_{\text{Neg}}$ contrast maps for different healthy subjects ($n=6$) with and without MT removal using different lineshapes. The $\text{GluCEST}_{\text{Neg}}$ contrast was also higher in GM compared with WM both before and after MT removal. The $\text{GluCEST}_{\text{Neg}}$ contrast decreases significantly ($p < 0.01$) after MT removal corresponding to all the lineshapes. $\text{GluCEST}_{\text{Neg}}$ or conventional $\text{GluCEST}_{\text{Neg}}$ showed contrast ratio of GM/WM = 1.28, whereas $\text{GluCEST}_{\text{Neg}}$ after MT removal showed contrast ratio 1.52, 1.79, 1.42, and 1.79 for Lorentzian, Gaussian, super-Lorentzian, and 6th degree polynomial respectively.

MRS Spectroscopy Results

Figure 5 a shows anatomy image of a representative *in vivo* healthy human volunteer at 7T with voxel location placed on GM (white rectangle) and WM (red rectangle) respectively. Two rectangular ROIs were placed over GM and WM (Fig. 5a). Figure 5 b, c shows ^1H MRS spectra obtained from the respective voxels on GM and WM. The glutamate concentration ratio in GM/WM from ^1H MRS was 1.6 ± 0.13 . Figure 5 d shows the $\text{GluCEST}_{\text{M0}}$ map before MT removal

and Fig. 5 e–h shows the $\text{GluCEST}_{\text{M0}}$ maps after MT removal using Lorentzian, Gaussian, super-Lorentzian, and 6th degree polynomial lineshape respectively of the same representative volunteer. Table 4 shows that the $\text{GluCEST}_{\text{Neg}}$ and $\text{GluCEST}_{\text{M0}}$ contrast GM/WM ratio after Lorentzian-based MT removal were closest to the glutamate concentration ratio in GM/WM as observed in the current spectroscopic study.

Interpolation for Super-Lorentzian Lineshape

Figure S2b–c (see ESM) shows that the super-Lorentzian lineshape creates a hump or pole at the offset frequency center of the MT pool when the lineshape is interpolated over the entire frequency range. To eliminate the hump, spline interpolation has been used (Fig. S2d, see ESM) over the frequency range of -14 to $+14$ ppm, instead of using fitted parameters to generate interpolation by super-Lorentzian lineshape.

Rat Tumor Model

Figure 6 shows $\text{GluCEST}_{\text{M0}}$ maps for a tumor-bearing rat before and after MT removal along with the anatomical

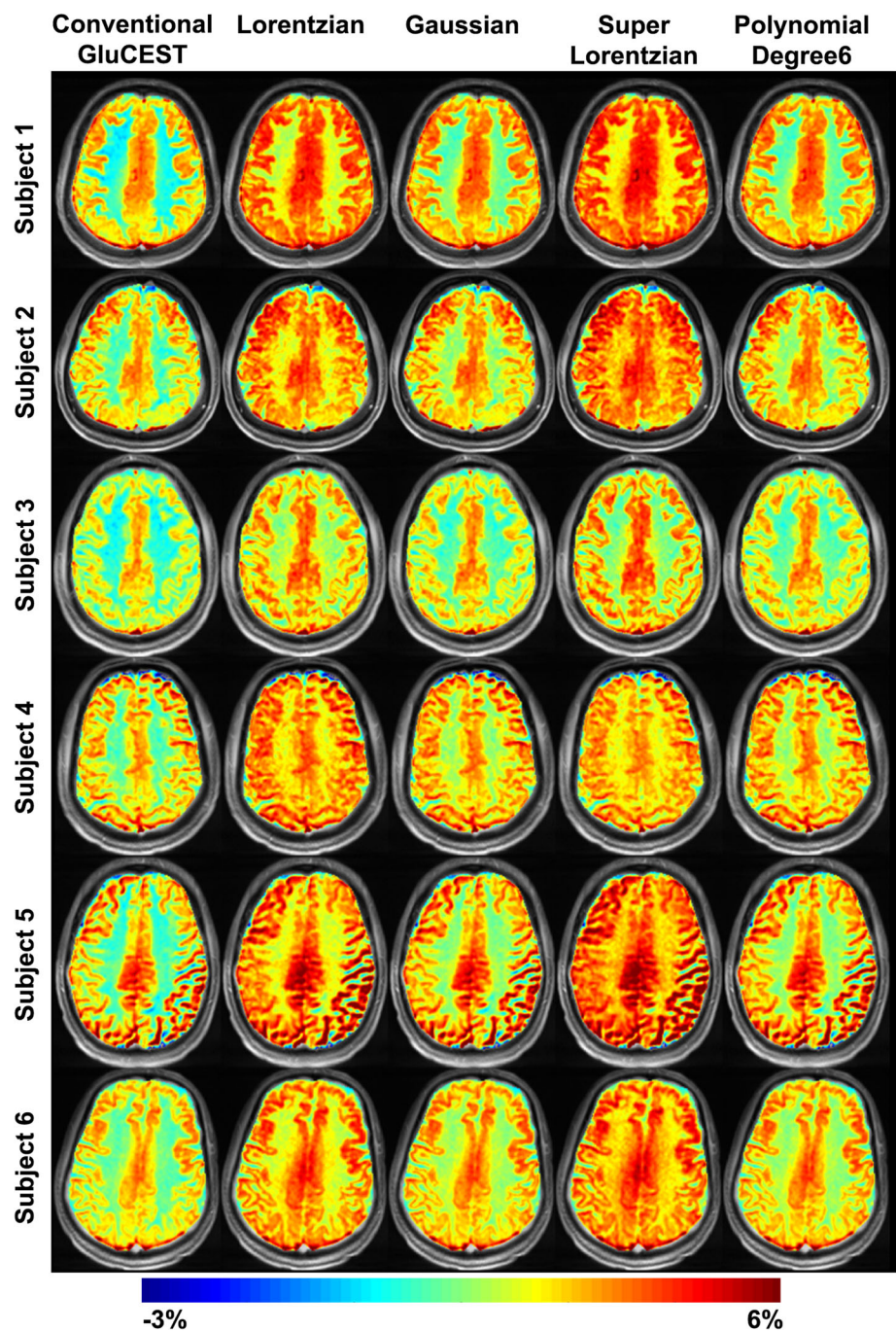


Fig. 3. GluCEST_{M0} contrast before and after MT removal for different healthy human subjects. The conventional GluCEST_{M0} contrast before MT removal for all subjects are shown in first column. The GluCEST_{M0} contrast after MT removal for all subjects using lineshapes to model MT components—(second column) Lorentzian, (third column) Gaussian, (fourth column) super-Lorentzian with spline interpolation, and (fifth column) 6th degree polynomial function lineshapes. There is an elevation in GluCEST_{M0} contrast after MT removal for all lineshapes. Color bars represent percentage GluCEST_{M0} contrast.

image. The conventional GluCEST_{M0} contrast is higher in the tumor region (8.59 %) compared with the contra-lateral region (4.71 %) (Fig. 6b). After MT removal using Lorentzian lineshape, GluCEST_{M0} contrast increases in whole brain region. Figure 6 c shows GluCEST_{M0} contrast was higher in the tumor region

(10.72 %) than the contra-lateral region (7.19 %) (Fig. 6c). The MTR value is higher in the tumor region (38.18 %) compared with the contra-lesion region (23.79 %) as shown in Fig. 6 d. Fig. S3 (see ESM) shows $\text{GluCEST}_{\text{Neg}}$ maps for a tumor-bearing rat before and after MT removal.

Table 2. GluCEST contrast with and without MT removal

Tissue	Lineshape	GluCEST _{M0} (%)	<i>p</i> value	GluCEST _{Neg} (%)	<i>p</i> value
GM	Before MT removal	3.17 ± 0.38		7.13 ± 0.87	
	Lorentzian	3.87 ± 0.35	0.003*	4.69 ± 0.44	< 0.001*
	Gaussian	3.46 ± 0.37	0.17	4.68 ± 0.48	< 0.001*
	Super-Lorentzian	4.17 ± 0.38	0.0003*	5.21 ± 0.48	< 0.001*
	Polynomial degree 6	3.53 ± 0.37	0.09	4.68 ± 0.48	< 0.001*
WM	Before MT removal	1.22 ± 0.34		5.54 ± 0.85	< 0.001*
	Lorentzian	2.64 ± 0.28	< 0.001*	3.08 ± 0.34	< 0.001*
	Gaussian	1.85 ± 0.32	0.004*	2.66 ± 0.42	< 0.001*
	Super-Lorentzian	3.14 ± 0.31	< 0.001*	3.66 ± 0.37	< 0.001*
	Polynomial degree 6	1.87 ± 0.33	0.003*	2.61 ± 0.42	< 0.001*

Asterisk represents significant difference; *WM* white matter; *GM* gray matter

Discussion

The current study focuses on modeling of semi-solid macromolecular MT pool using partial Z-spectral fitting with appropriate lineshape and evaluates GluCEST contrast after removal of MT pool from Z-spectra. To the best of our knowledge, this is the first *in vivo* study on human brain at 7T related to removal of MT component from Z-spectra before computing GluCEST contrast. This approach results in removal of MT asymmetry contribution to GluCEST contrast.

Z-spectra is a combination of several effects from DS, MT, and CEST effects from different metabolites/molecules and rNOE pools. The effect of the DS does not make the Z-spectra asymmetric; rather, it is the MT effects around the water resonance that makes Z-spectra asymmetric [23]. The amount of asymmetry in MT effect depends upon tissue under consideration, saturation parameters, and offset frequency. Moreover, other factors that generate asymmetric Z-spectra are CEST and rNOE effects. Due to the presence of inherent asymmetry in the Z-spectra, computation of the desired CEST only maps is not feasible [35]. The MT has detrimental effects on Z-spectra and hence causes a systematic error in the quantification of GluCEST [47]. Thus, it is quite important to remove MT effect from Z-spectra while computing GluCEST. For GluCEST imaging, Z-spectral data used in the current study were acquired from human brain at high B_{1rms} (2.9 μ T) and with shorter duration (800 ms). High saturation powers attenuates/suppresses the rNOE contributions to Z-spectra in both WM and GM, thus

enhancing the detection of CEST contrast. The rNOE decrease with increase in B_1 power level which was reported by several studies [20, 48, 49]. We assumed that contribution of rNOE effect in Z-spectra is negligible at B_{1rms} of 2.9 μ T and short duration. At these saturation parameters, APT effect is also quite small. Contribution of other CEST effects from creatine, glucose, myoinositol, *etc.* is also expected to be low due to well separation of offset frequencies. Thus, the major confounding effect to GluCEST contrast is asymmetry from MT asymmetry. Notably, with the removal of MT effect from Z-spectra, specificity of GluCEST to glutamate concentration can be increased. Therefore, in our study, we showed the feasibility of MT effect removal from GluCEST contrast thereby improving the specificity of GluCEST to glutamate concentrations (Figs. 3, 4, and 5). The saturation parameters used for the rat study was different from that of human study. These parameters provide higher GluCEST effect; however, it is not feasible to use these for human study due to SAR effect and hardware constraints.

The current study compares MT modeling using Lorentzian, Gaussian, and super-Lorentzian lineshapes along with polynomial functions. Many studies [33] showed that super-Lorentzian lineshape has been successfully used to model MT in many biological tissues. Our results were consistent with previous studies [33, 39] which showed fitting of Z-spectra at far-off resonance to model semi-solid MT is best using super-Lorentzian lineshape. One key difference of our study to Morrison et al. [33] is that we have used these lineshapes to directly fit Z-spectra rather

Table 3. Initial parameters to fit Z-spectra with its lower and upper bounds for MT modeling

Tissue	T_{2m} (μ s)	w_0 (ppm)	A (A.U.)
Lower bound	1	−3	0
Initial value	20	−2.4	0.5
Upper bound	30	−1	1

A—scaling factor; A.U.—arbitrary unit; T_{2m} is time constant and is not to be confused with transverse relaxation time (T_2) of the MT pool [39]. This T_{2m} characterizes the linewidth of the solid macromolecular MT pool [33]. w_0 is offset frequency of the MT pool which varies across tissues from −1 to −2.5 ppm as reported in various studies [44, 45]

Table 4. Tabulating GluCEST contrast before and after MT removal using different lineshapes. The glutamate concentration ratio in GM/WM was 1.6 ± 0.13 as observed from spectroscopic study. The GluCEST contrast based upon Lorentzian lineshape, after removal of MT, only provides contrast ratio closest to spectroscopic ratio

	Lineshape	GluCEST _{M0} (GM/WM)	GluCEST _{Neg} (GM/WM)
Before MT removal		2.84	1.32
After MT removal	Lorentzian	1.47	1.54
	Gaussian	1.94	1.71
	Super-Lorentzian	1.34	1.43
	Polynomial degree 6	1.97	1.84

GM gray matter; WM white matter

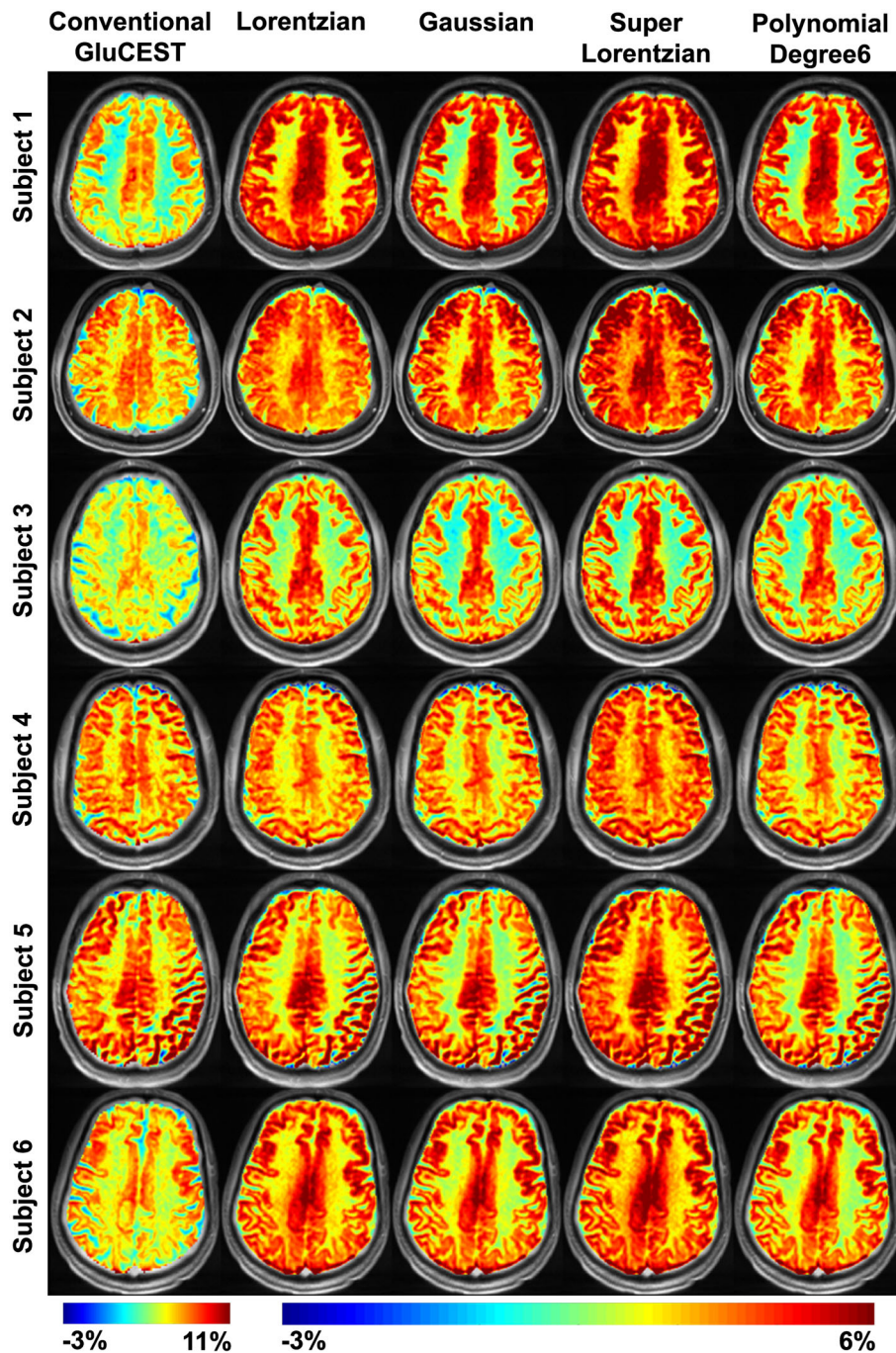


Fig. 4. $\text{GluCEST}_{\text{Neg}}$ contrast before and after MT removal for different healthy human subjects. The conventional $\text{GluCEST}_{\text{Neg}}$ contrast before MT removal for all subjects are shown in first column. The $\text{GluCEST}_{\text{Neg}}$ contrast after MT removal for all subjects using lineshapes to model MT components—(second column) Lorentzian, (third column) Gaussian, (fourth column) super-Lorentzian with spline interpolation, and (fifth column) 6th degree polynomial functions. The $\text{GluCEST}_{\text{Neg}}$ contrast decreases after MT removal for all lineshapes. Color bars represents percentage $\text{GluCEST}_{\text{Neg}}$ contrast. For better visualization, the conventional $\text{GluCEST}_{\text{Neg}}$ contrast (first column) has one scale range and rest of the columns have different scale range.

than using these as lineshape of macromolecular spectrum. In addition, we have further observed a reliable fitting with lowest average residual errors per point utilizing super-Lorentzian lineshape to characterize the MT spectrum. This was also consistent with previous study [33] reporting

highest quality of fit for super-Lorentzian followed by Lorentzian and Gaussian lineshape. One point to note is that variation in σ is less irrespective of the lineshape used and reasonably good fits were obtained with all the four lineshapes. The current MT removal method fits the partial

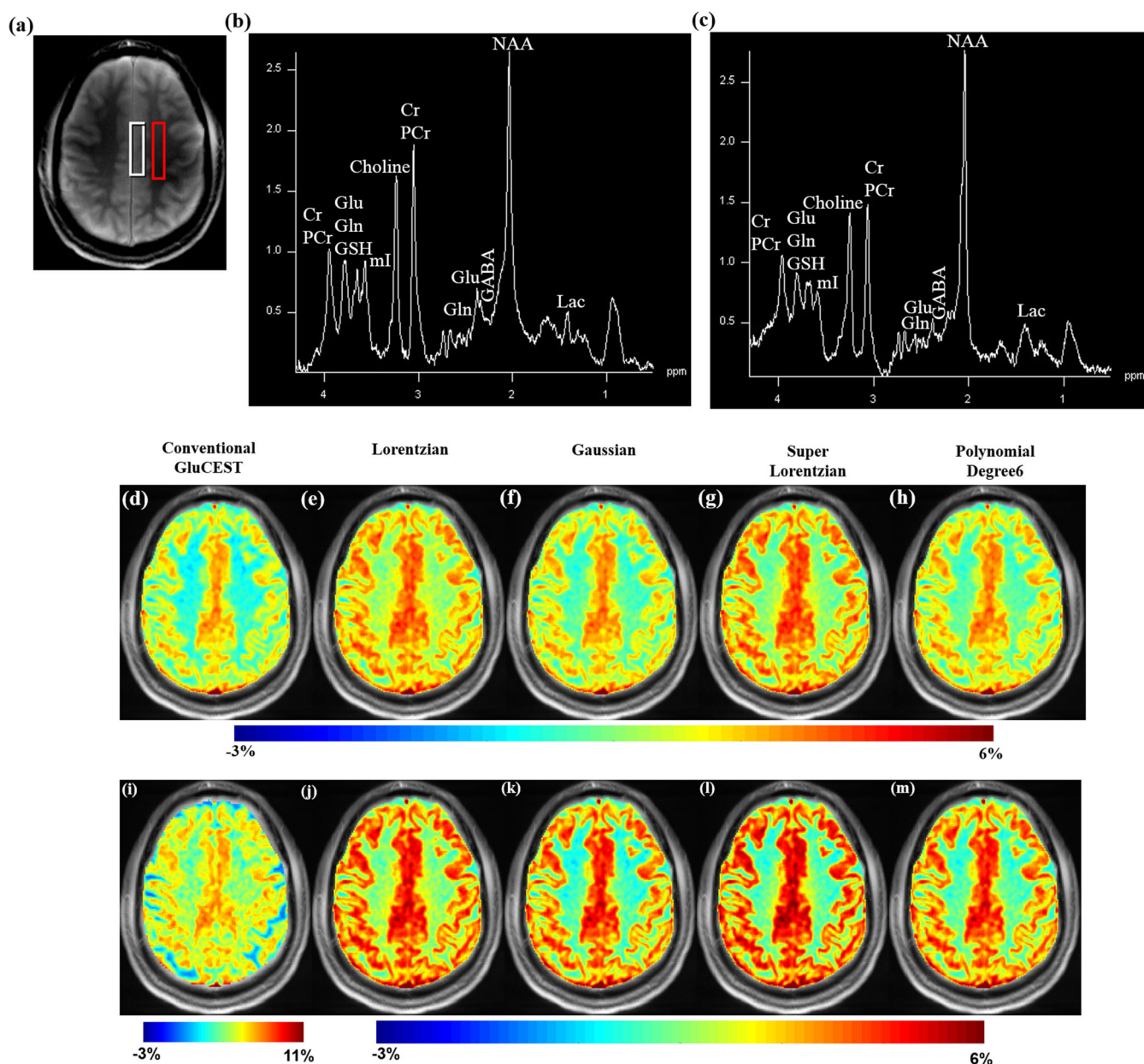


Fig.5. **a** Anatomy image of a representative *in vivo* healthy human volunteer at 7T with voxel location placed on GM (white rectangle) and WM (red rectangle) respectively. ^1H MRS spectra obtained from the respective voxels on **b** GM and **c** WM. **d** $\text{GluCEST}_{\text{M0}}$ map before MT removal. $\text{GluCEST}_{\text{M0}}$ maps after MT removal using **e** Lorentzian, **f** Gaussian, **g** super-Lorentzian, and **h** 6th degree polynomial lineshapes respectively. Abbreviation: WM, white matter; GM, gray matter.

Z-spectra acquired at 7T over ± 14 to ± 100 ppm offset frequency range assuming that only MT effects dominates over this range and there is no CEST, DS, and rNOE effects. Bloch McConnell simulations need to be further carried out considering multi-pool to show that there is only MT effect over this offset frequency range. The contribution of MT to the Z-spectra varies with use of different lineshapes to fit MT component. Presumably, the quantity of removal of MT from Z-spectra varies depending upon the selection of the lineshape. Hence, the question comes up regarding the appropriate lineshape to be used for the removal of correct

quantity of MT from Z-spectra. To resolve this issue, we correlated the ratio of GluCEST contrast in GM and WM (GM/WM ratio) before and after MT removal with ratio of glutamate concentration in respective tissues. Based upon the spectroscopic measurement, glutamate exhibits a ratio of ~ 1.6 between GM and WM [8, 10]. The spectroscopic result of the current study also showed that the glutamate concentration in GM/WM is of 1.6 ± 0.13 . The $\text{GluCEST}_{\text{Neg}}$ contrast ratio between GM and WM was reported to be ~ 1.6 in both rat and human brain [8]. In the current study, the conventional $\text{GluCEST}_{\text{Neg}}$ contrast showed a ratio of 1.3.

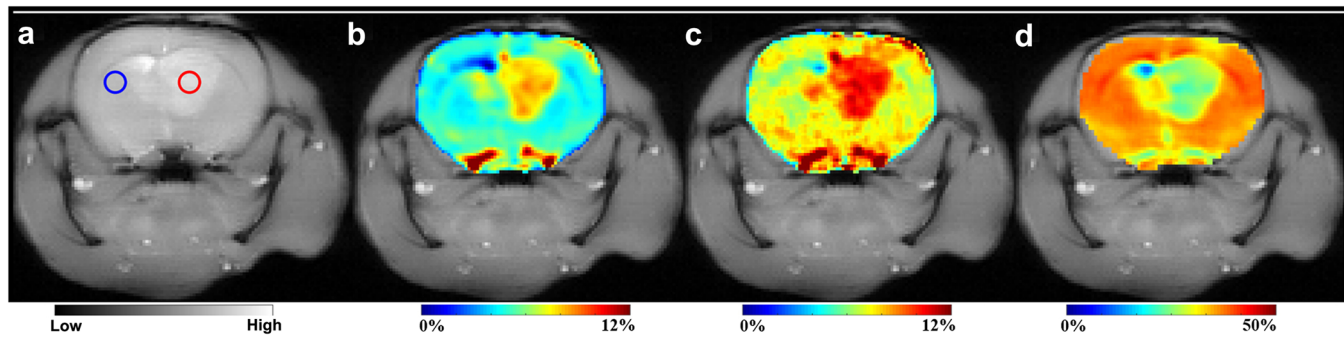


Fig. 6. The conventional GluCEST_{M0} maps of a rat brain with tumor: **a** anatomic brain image of unsaturated signal at 100 ppm (M_0) demonstrating tumor region (red circle) and contra-lesion normal appearing region (blue circle); **b** conventional GluCEST_{M0} map; **c** GluCEST_{M0} map after removal of MT effect using Lorentzian lineshape; **d** MTR map at 20 ppm. Abbreviation: MT, magnetization transfer; ROI, regions of interest.

This may be due to the use of different saturation parameters and partial volume effect. On the other hand, conventional GluCEST_{M0} show a ratio of 2.6, which is quite high. This high ratio is clearly showing that MT and DS (spillover effect) effect still influencing GluCEST_{M0} . Thus, it manifests that due to asymmetry in MT effect, conventional GluCEST_{M0} and GluCEST_{Neg} may not provide true measurement of GluCEST contrast and may not preserve glutamate concentration ratio between GM and WM. Thus, MT removal is important. In the proposed approach for GluCEST contrast computation after removal of MT, GluCEST_{M0} (Lorentzian based MT fitting) provided GM/WM ratio = 1.46, which is close to glutamate concentration ratio between GM and WM tissues. Similarly, GluCEST_{Neg} (Lorentzian based MT fitting) showed a ratio of 1.52 which provided closest glutamate concentration in GM/WM ratio. GluCEST_{M0} and GluCEST_{Neg} contrast after MT removal using super-Lorentzian, Gaussian, and polynomial lineshapes provide large differences for the GM/WM compared to spectroscopic-based glutamate concentrations. It is noteworthy to mention that the quantity of MT removed from Z-spectra is overestimated using super-Lorentzian (Fig. 1k), while Gaussian (Fig. 1j) and polynomial (Fig. 1l) fitted MT provides underestimation of MT contribution to Z-spectra. The quantity of MT removed from Z-spectra is appropriate using Lorentzian lineshape (Fig. 1i) since it can preserve contrast ratio.

Potential contribution of the asymmetry in MT asymmetry to GluCEST leads to an underestimation of actual GluCEST_{M0} contrast [8]. After removal of the confounding MT, the GluCEST_{M0} contrast increases in both GM and WM. There is a significant difference ($p < 0.05$) between GluCEST_{M0} contrast before and after MT removal using both super-Lorentzian and Lorentzian lineshape in both the tissues. Among all lineshapes, Lorentzian could preserve the contrast ratio between WM and GM in GluCEST_{M0} contrast after MT removal and provide the true estimate of glutamate concentration. The increase of GluCEST_{M0} contrast is higher in WM compared to GM due to the fact that MT is more in WM than in GM [10, 43]. Thus, the quantity of MT removal

is more in WM compared to GM. Our results are therefore consistent with the previous published studies [10, 43] which reported higher GluCEST contrast in GM than in WM due to higher glutamate concentrations in GM compared to WM. Moreover, in GluCEST_{Neg} , normalization changes the dynamic range [19]. Our results are consistent with this. The conventional GluCEST_{Neg} contrast was higher in both GM and WM. After MT removal, GluCEST_{Neg} contrast decreases significantly ($p < 0.01$) in both tissues for all the lineshapes. Among all lineshapes, Lorentzian could preserve the contrast ratio between WM and GM in GluCEST_{Neg} contrast after removal of MT and provide the true estimation of glutamate concentration.

Previous studies on MRS [50, 51] reported that tumor regions show increased levels of glutamate and glutamine (Glx) concentration compared with normal region. However, increase in glutamate concentration can vary depending upon tumor type and is not well reported. Additional factors could also contribute to elevate GluCEST values in tumor such as lower pH in tumor [52]. Cai et al. [10] showed that with the decrease in pH for stroke/ischemic rat brain, the GluCEST contrast increases substantially. Moreover, there is an elevation in protein/peptide concentration, which might also provide small contribution to GluCEST contrast [48, 49]. However, further systematic studies are required to compute contribution of change in concentration, pH, and mobile proteins/peptides to GluCEST contrast.

It is noteworthy to discuss that the age of the subjects used in the study has overall effect on the glutamate concentrations and MT effects. A previously reported study [53] has shown age-dependent alteration of glutamate concentrations in human brain. Over the course of aging, glutamate concentration gets depleted in various regions of brain. The cohort included in the current study has variability in age and thus possess inter-subject variability in glutamate contrast. The variation in the distribution of GluCEST contrast in WM and GM is indeed suggestive of changes of glutamate with advancing age. Moreover, MT changes significantly with age due to degeneration of myelin sheath [54]. For obtaining an accurate correlation between

MT and age, we need to acquire more data on subjects with varying age. The sample size of our study is quite small to make any conclusion. Still, it can be said that the MT removal plays a vital role in accurately measuring GluCEST in case of subjects with significant difference in age. Further studies are needed to look into the estimated MT removed age-related changes in GluCEST maps.

Preliminary study has been carried out in this article to show preclinical application and importance of removal of MT component. It has been well reported that GluCEST contrast increases in tumor region compared to contra-lateral normal region [10], while MT decreases in tumors [41]. Therefore, low MT results in artificial elevation in GluCEST contrast in tumor region. Thus, MT removal is highly recommended for true estimation of GluCEST contrast. In the current study, MT uncorrected GluCEST_{M0} contrast is 82 % higher in tumor region than in the normal region. Due to MT, GluCEST shows false elevation of contrast in tumor region. After MT removal, GluCEST_{M0} contrast is 50 % higher in tumor region compared with normal region. Thus, true GluCEST contrast can be estimated after removal of MT component.

It is also worth discussing some of the potential clinical applications where MT correction for GluCEST contrast is appropriate. Patients with evidence of Alzheimer's disease (AD) have shown a decrease in glutamate concentration [16]. Moreover, a decrease in the MT asymmetry contrast has been reported in AD patients which corresponds to myelin abnormalities in WM [55]. Therefore, a decrease in MT asymmetry contrast will result in an increase in GluCEST contrast due to its confounding effect. On the other hand, decrease in glutamate concentration will result in a decrease in GluCEST contrast. Therefore, for AD patients, conventional GluCEST approach might not be appropriate as any decrease in the MT asymmetry contrast might suppress a decrease in glutamate concentration. However, using our proposed method, MT contribution can be removed and hence any change in glutamate concentration can be detected. In such scenarios, the MT removal approach plays a vital role. However, a systematic study is required to validate this claim. There are some other clinical conditions like multiple sclerosis (MS) where both GluCEST and MT asymmetry increases and in such scenarios, MT removal approach plays a major role in quantification of GluCEST contrast.

The current study has following limitations along with future directions. Firstly, the contribution of rNOE to Z-spectra and GluCEST was assumed to be negligible at the saturation parameters used for GluCEST imaging. Further investigation needs to be carried out for estimation of exact amount of rNOE contribution to GluCEST and its subsequent removal from GluCEST. Secondly, we could not acquire spectroscopy data for the reported rat tumor data. Moreover, only three human subjects were scanned for spectroscopy data; other subjects could not be scanned for spectroscopy data as they have moved to different

city/country. Additionally, there was a gap of 1 year between CEST data and spectroscopy data acquisitions. However, these were healthy young volunteers, so we do not expect changes in glutamate concentration within a year. Lastly, acquisition of additional data for modeling MT effect requires longer scan times. Future studies can be carried out to speed up data acquisition or using alternative approaches based upon machine learning to model MT using only few off-resonance frequency data.

Conclusions

Conventional GluCEST_{M0} and GluCEST_{Neg} showed underestimation and overestimation of contrast due to confounding MT effect. The study evaluated the feasibility of estimating GluCEST_{M0} and GluCEST_{Neg} contrast by removal of MT effect from Z-spectra using different line-shapes for MT component. The proposed method with Lorentzian lineshape provided significant differences in contrast before and after MT removal and could also preserve the actual glutamate contrast ratio between GM and WM as measured with the spectroscopic results. This approach increases specificity of GluCEST to glutamate concentration improving GluCEST estimation which might have potential in diagnosis, monitoring, and treatment of diseases.

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Compliance with Ethical Standards

Conflict of Interest

The authors declare that they have no conflict of interest.

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